

HEALTHCARE ANALYTICS · DATA STORY

Who Goes to the Doctor? A Data Story

Healthcare Analytics for Doctor Visits · Australian Health Survey · n = 5,190 respondents · 13 variables

5,190

TOTAL PATIENTS

79.8%

ZERO-VISIT RATE

0.302

AVG VISITS / PATIENT

13

VARIABLES

EXECUTIVE SUMMARY

The Big Picture

This report analyses the **Australian Health Survey** dataset of 5,190 individuals, examining the factors that influence how often people visit a doctor. The 13 variables span demographics (age, gender, income), health status (illness severity, chronic conditions, self-rated health), functional impairment (activity-reduced days), and three distinct insurance coverage types.

The single most striking feature of the data is its **zero-inflated distribution** — 79.8% of respondents recorded zero doctor visits. This reflects a genuine two-part behavioural story: *whether* people visit, and *how often* they visit once they cross that threshold.

5,190

TOTAL PATIENTS

79.8%

ZERO-VISIT RATE

0.302

AVG VISITS

44.3%PRIVATE
INSURANCE**40.3%**NON-LIMITING
CHRONIC**11.7%**LIMITING
CHRONIC

SECTION 1

The Paradox of Non-Utilisation

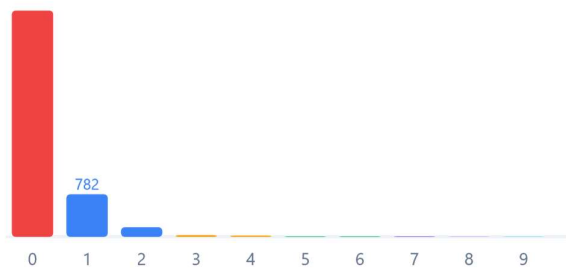
In this dataset of 5,190 Australians, the most common number of doctor visits is **zero**. This zero inflation carries profound implications — it means the question shifts from "how many times did people visit?" to **what crosses the threshold that separates zero visits from any visit at all?**

79.8% of patients — 4,141 of 5,190 — recorded zero doctor visits.

The modal response in this healthcare dataset is no healthcare at all.

DOCTOR VISITS DISTRIBUTION

Red = zero visits (4,141). Blue = 1–2 visits. Yellow = 3–4. The tail is extremely thin.



CUMULATIVE % OF PATIENTS (CDF)

94.9% of patients have ≤ 1 visit. Only 5.1% have 2+ visits — the "high-utilisation" minority.



Among those who did visit, the mean was **1.04 visits**. Zero-inflation almost certainly reflects two populations: those genuinely not needing care, and a latent group who needed care but did not access it — due to cost, geography, or cultural barriers.

SECTION 2

When Illness Strikes, People Visit

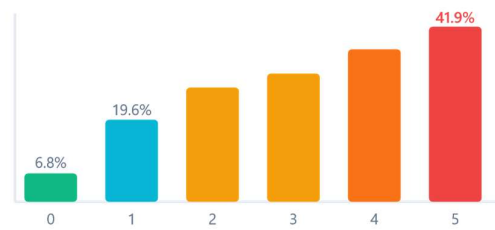
Illness severity is the single clearest predictor. Moving from no illness to maximum severity produces a **10× increase** in average visits — from 0.079 to 0.814.

AVG VISITS BY ILLNESS SEVERITY

Near-linear escalation from **0.079** at severity 0 to **0.814** at severity 5.

**% PATIENTS WHO VISITED — BY SEVERITY**

Only **6.8%** of severity-0 patients visited vs **41.9%** at severity 5 — a 6× probability increase.



Illness Severity	Avg Doctor Visits	% Who Visited
0 — None	0.079	6.8%
1 — Mild	0.295	19.6%
2 — Moderate	0.404	27.3%
3 — Significant	0.421	30.6%
4 — Severe	0.577	36.5%
5 — Very Ill	0.814	41.9%

Even at severity 5, **more than half of respondents still recorded zero visits** — illness severity alone is insufficient to guarantee utilisation. Other barriers intervene even when health need is acute.

SECTION 3

The Chronic Condition Burden

The dataset distinguishes between **non-limiting chronic conditions** (*nchronic*: managed conditions like controlled hypertension) and **limiting chronic conditions** (*lchronic*: conditions that restrict daily activities).

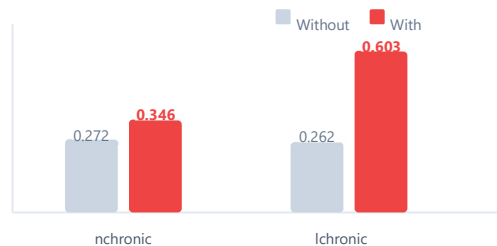
Their impact on visit rates differs dramatically.

A limiting chronic condition increases doctor visits by 130%.

Non-limiting conditions add a significant but more modest +27% uplift.

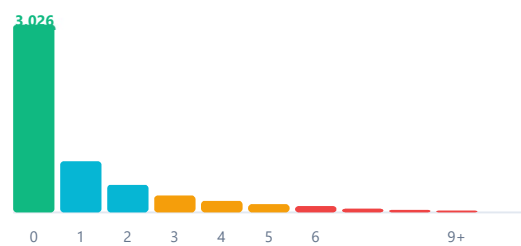
CHRONIC CONDITION IMPACT ON AVG VISITS

Without vs with condition. Limiting chronic: **+130%**. Non-limiting: **+27%**.



HEALTH SCORE DISTRIBUTION (0 = BEST)

58.3% report score 0 (excellent health). Higher scores are rare but correlate with elevated visits.



Chronic disease prevalence is a **primary driver of healthcare demand forecasting**. In an ageing population, the proportion with limiting chronic conditions will only grow — compounding pressure on primary and specialist care.

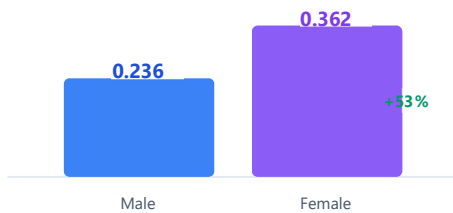
SECTION 4

Gender, Age & the Social Gradient

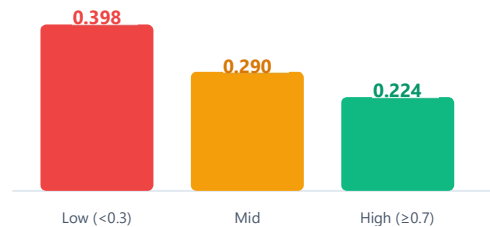
Gender, age, and income each independently shape utilisation. Together they define a structural gradient: **older, lower-income women are the highest-utilisation subgroup**; younger, higher-income men are the lowest.

AVG VISITS BY GENDER

Females: **0.362** vs Males: **0.236** — a consistent **53% gap** across all subgroups.

**AVG VISITS BY INCOME BRACKET**

Low-income: **0.398**. Mid: **0.290**. High: **0.224**. A clear inverse social gradient.

**AVG VISITS & PATIENT COUNT BY AGE GROUP**

The oldest cohort (0.72 ≈ 65+) averages **0.483 visits** — 2.4× more than the youngest groups (0.201). Age-upward trend with a mid-life dip.



Low-income: 0.398 visits · Mid-income: 0.290 · High-income: 0.224

Healthcare utilisation decreases as income increases — need-driven, not discretionary.

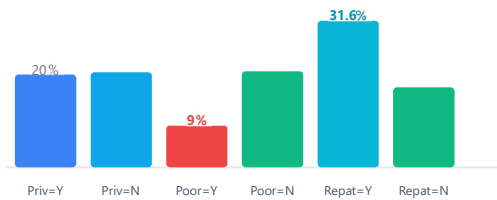
SECTION 5

The Insurance Paradox

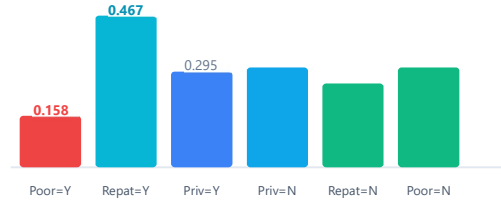
Insurance findings are counterintuitive. **Freerepat** cardholders (veterans/families) have the highest visit rate (31.6%, avg 0.467). **Freepoor** cardholders — the most financially vulnerable — have the *lowest* visit rate (9.0%, avg 0.158). **Private insurance** shows essentially no effect (0.295 vs 0.307).

% PATIENTS VISITING — BY INSURANCE TYPE

Freerepat: **31.6%**. Freepoor: **9.0%** — the most vulnerable group visits least.

**AVG VISITS BY INSURANCE SCHEME**

Freerepat: **0.467**. Freepoor: **0.158**. Private: **no meaningful edge**.



The **freepoor paradox** is the starkest illustration in this dataset: removing the financial barrier alone is clearly not sufficient. Geographic distance, cultural factors, distrust of institutions, or navigating a healthcare system while managing poverty all play a role. Non-financial access barriers must be addressed alongside cost coverage.

SECTION 6

Functional Impairment as a Warning Sign

Activity-reduced days — days unable to perform normal activities due to illness — provide a strong objective predictor of visits. From 0.184 visits at 0 reduced days to 2.0 at 10 days, the trend is strongly positive.

ACTIVITY-REDUCED DAYS VS AVG DOCTOR VISITS

Near-monotonic rise from **0.184** (0 days) to **2.0** (10 days). Spike at 11 days (5.0 visits) is a small-n outlier.

n=2 at day 11



Activity-reduced days are a powerful leading indicator of healthcare demand. A monitoring system tracking activity limitation in real time could serve

as an early-warning signal for individuals at risk of high utilisation, enabling proactive rather than reactive care delivery.

SECTION 7

Modelling Recommendations

Standard linear regression is **entirely inappropriate** for this data. Doctor visits are non-negative integer counts with overwhelming zero-inflation. Three frameworks are recommended:

1. Zero-Inflated Poisson (ZIP) — mixture of structural-zero logistic model + Poisson count model.

Separates *who visits* from *how often*.

2. Zero-Inflated Negative Binomial (ZINB) — preferred when count component is overdispersed (variance > mean), which is likely here.

3. Hurdle Models — two-part model; first part: binary (any visit), second: count given ≥ 1 visit. Simpler to interpret.

Rank	Variable	Role
1	Illness severity (illness)	Strongest single predictor; drives both hurdle and count components
2	Limiting chronic (lchronic)	+130% visit uplift; large effect size
3	Activity-reduced days (reduced)	Objective impairment; strong count-component predictor
4	Age (age)	Monotonic increase; structurally important
5	Gender	Consistent 53% female uplift; robust across subgroups
6	Income	Inverse gradient; important for equity analysis
7	Non-limiting chronic (nchronic)	Modest +27% effect; important control variable

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SUMMARY Key

Findings at a Glance

● Zero-Inflated Pattern

79.8% zero visits. ZIP/ZINB or hurdle models required.

● Illness is #1 Driver

0→5 severity: 0.079→0.814 visits (10×). 6× visit probability increase.

● Chronic Compounds

lchronic +130%. nchronic +27%. Compounding peak: 0.852 visits.

● Gender Gap

Females 53% more visits. Consistent across all subgroups.

● Income Gradient

Low income: 0.398; High: 0.224. 63% more likely to visit if low-income.

● Age Rises

Oldest cohort: 0.483 visits — 2.4× youngest (0.201). Ageing = rising demand.

● Activity Signal

0 days → 0.184; 10 days → 2.0 visits. Strong early-warning proxy.

● Freepoor Paradox

Most vulnerable: lowest visit rate (9%). Non-financial barriers persist.

Finding	Evidence	Implication
Zero-inflated distribution	79.8% (4,141/5,190) report zero visits	ZIP/ZINB or hurdle models required
Illness severity drives utilisation	0.079 → 0.814 visits (10× increase)	Perceived health is the primary care trigger
Limiting chronic conditions	lchronic +130%; nchronic +27%	Chronic burden must drive resource allocation
Gender and age gradient	Female +53%; 65+ 2.4× more than youngest	Services should target older female populations
Inverse income gradient	Low 0.398, mid 0.290, high 0.224	Lower SES groups carry higher disease burden
Freepoor paradox	Lowest utilisation (9%) despite maximum need	Address non-financial access barriers
Functional impairment predictor	0 days → 0.184; 10 days → 2.0 visits	Activity monitoring as early-warning signal

CONCLUSION

From Data to Policy

This dataset tells a story of a healthcare system under two simultaneous pressures: **over-utilisation by the most ill** (the small minority with chronic conditions, severe illness, and functional impairment) and **underutilisation by the most vulnerable** (freepoor cardholders, low-income groups, and structurally

absent zero-visitors who likely need care but are not getting it). The average visit rate of 0.302 obscures this bimodal reality — which is precisely why zero-inflated statistical methods are essential for sound healthcare planning.

The policy implications are clear. Reducing zero-visit rates among high-risk groups requires interventions that go beyond making healthcare free. The freepoor paradox is the starkest illustration: cost coverage exists, but utilisation remains the lowest of any group. Outreach, health literacy, community health workers, and reducing structural barriers must accompany financial subsidies. The data demands a shift from simply *providing* healthcare to actively *connecting* people to it — meeting patients where they are, not waiting for them to arrive.

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